Thoughts And Rambles On Probability And Statistics

Problem Set 4: Inequalities

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These are the end of chapter problems for Chapter 4 of the notes. Equation numbers refer to the full notes.

1. Show that, for any positive random variable X,

$$\mathbb{E}\left[\max\left\{X, \frac{1}{X}\right\}\right] \geq 1$$

2. Let $X \sim \text{Poisson}(\lambda)$. Use Chebyshev's inequality to show that $\mathbb{P}(X \geq 2\lambda) \leq 1/\lambda$.

3. Prove Mill's inequality Let $Z \sim N(0,1)$. Then,

$$\mathbb{P}(|Z| > t) \leq \sqrt{\frac{2}{\pi}} \frac{e^{-t^2/2}}{t}$$

4. Prove that, for any random variable X, the minimum of $E(X-a)^2$ is attained for $a = EX$.

5. Prove that, for any random variable X, the minimum of $E|X-a|$ is attained for $a = \text{med}(X)$.

6. Let $X_1, \ldots, X_n \sim \text{Bernoulli}(p)$ and $\bar{X}_n = n^{-1}\sum_{i=1}^n X_i$. Bound $\mathbb{P}\left(|\ \bar{X}_n - \ p\ | > \epsilon\right)$ using Chebyshev's inequality and using Hoeffding's inequality. Show that, when n is large, the bound from Hoeffding's inequality is smaller than the bound from Chebyshev's inequality.